07. Model Training & Hyperparameter Tuning

Exact training flow, parameters, and what was not tuned.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Notebook fitting flow

- Create 15 cleaned features and one-hot encode Education and Living_With to 18 columns.
- StandardScaler is fit on the full encoded matrix.
- PCA(n_components=3) is fit on the full scaled matrix.
- Fit K-Means for K=1..10 and collect inertia; KneeLocator reports K=4.
- Compute silhouette scores for K=2..10 and plot them.
- Fit K-Means with K=4 for a 3D visualisation.
- Fit AgglomerativeClustering(n_clusters=4, linkage='ward') and use those labels for the cluster profile.

Hyperparameters

Component	Parameter	Value
StandardScaler	Default settings	Mean 0, variance 1 per feature
PCA	n_components	3
K-Means sweep	n_clusters	1 through 10
K-Means	random_state	42
Final K-Means	n_clusters	4
Agglomerative	n_clusters	4
Agglomerative	linkage	ward

What is missing

- No systematic search over PCA dimensions, distance metrics, linkage methods or K-Means initialisation settings is reported.
- The notebook does not report n_init explicitly; behaviour can vary across scikit-learn versions.
- No resampling/stability analysis is performed.
- No frozen preprocessing pipeline or method for assigning new customers is saved in the notebook.
- No comparison of cluster quality in original 18-dimensional space versus 3D PCA space is documented.

Recommended experiment matrix

- Compare 2–6 PCA components and no-PCA baseline.
- Compare K-Means, Ward agglomerative and Gaussian mixtures using the same preprocessed feature set.
- For each K, record inertia, silhouette, Davies–Bouldin, Calinski–Harabasz and minimum cluster size.
- Repeat K-Means across multiple random seeds; measure assignment stability after aligning cluster IDs.
- Compare profiles before and after excluding Response to detect campaign-outcome leakage effects.

Interview detail

K-Means is used for the elbow/silhouette exploration and a visual cluster fit. The notebook then assigns final cluster labels with Ward AgglomerativeClustering. State this distinction clearly if asked which model generated the final segment profiles.